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Theory and methodology of science with application (Kungliga Tekniska Högskolan)



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THEORY AND METHODOLOGY OF SCIENCE

1. SCIENTIFIC KNOWLEDGE

Method: An empirical practice within science, such as observational studies, field experiments or model studies.

Methodology: The systematic assessment and justification of method choice.

Proposition: The information expressed in a statement or a claim. "Snow is white" and "Snö är vitt" contains the same proposition, but they are two different statements.

Lexical definition: Definition that intends to capture common usage of a term.

Stipulative definition: Definition made for a specific purpose in a specific context.

Typical goals of science: Prediction, explanation and design.

Prediction: Knowing that an event will occur at a future time.

Explanation: Knowing what causes produced a certain event.

Design: Being able to design/come up with an artefact that satisfies certain desired functions.

How to choose a method? Justification of method choice:

a)Conventional view of methodology: following the established convention, what your peers/everyone else does, the "common practice" in the field. The question of method choice does not arise. Problems: It is hard to discover a mistake if the conventional method contributes to it somehow. Also interdisciplinary collaboration is harder if the conventional methods are different in each discipline.

b)The best-results view of methodology: choosing what gives the best/desired results, what serves a purpose best. Problem: vagueness, sometimes it is not possible to know which is the specific purpose we want to achieve (long-term plans). Also, who decides which one is the best result? Best result for who?

c)Epistemic tool view of methodology: choosing what one has more reasons to believe is the best tool for satisfying the goal. Goal need to be sufficiently specified. If the goal remains unclear it is outcome-oriented. It is a version of instrumental reasoning.

Instrumental reasoning: Providing justification to fulfill a clearly stated goal. It is used in everyday decisions (which degree to study or what to have for dinner).

The classical definition of knowledge: True, justified belief (Plato). "K knows P if: (1) K believes P to be true; (2) K has justification for P to be true; (3) P is in fact true".

Belief: A state of mind where a person considers a proposition to be true. It comes in degrees and so does knowledge. Both certain beliefs about false claims (e.g: sun revolves around Earth) and uncertain beliefs about true claims (e.g: getting 50% heads when flipping a coin several times) are possible.

Uncertainity: A belief is uncertain if it is not absolutely justified. Acknowledging uncertainty in one's knowledge is important in science, even our best theory could be wrong.

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Justification: reasons that can be put forward in defense of believing a certain proposition to be true. Not all relevant reasons for beliefs are justification for scientific knowledge (e.g: wishful thinking or religious convictions).

Weak internalism of justification: one is justified even if one does not have evidence or reasons (justifiers) in hand, it is enough if one could get them if invested sufficient effort (e.g: history book for historical event knowledge).

Truth: True statements describe part of how things really are.

Scientific knowledge for instrumentalist: They argue that creating a theory is just to rearrange and order observational reports, thus theories are never strictly true or false, entities it proposes are conceptual tools rather than something that exist.

Scientific knowledge for realists: They argue that theories are either true or false. The entities it proposes either exist or don't exist.

2. SCIENTIFIC INFERENCE

Inference: reaching a conclusion from a set of premises. Starting from something known and going to something else, still unknown.

Premise: A statement in an argument that justifies a conclusion.

Conclusion: A statement that follow logically from premises.

Conditional claim: claim saying that if a number of assumptions are true, then the conclusion is true. Claim involving the logical operator "if".

Inductive inference: Inference in which the premises support the conclusion but does not guarantee its truth. We observe a certain number of objects and then draw conclusions about that kind of objects. Examples of this are direct inferences, generalizations and projections. They are ampliative and, therefore fallible.

Deductive inference: Inference in which true premises necessitate the truth of a conclusion. We start in a general assumption and conclude something particular. They often comply conditional claims. Example of this are modus ponens and modus tollens. All the information is already contained in the premise, they are explicative and, therefore, truth-preserving.

Ampliative: Inferences that go beyond what it is stated in the premises. Inductive inference are ampliative.

Explicative: Inferences that do not go beyond what it is implicitly stated in the premises. Deductive inferences are explicative.

Truth preservation: The conclusion must be truth if the premises are true.

Fallibility: The conclusion can be false even if the premises are true.

Direct inference: Inductive inference from a sample to a population (If 1/3 of balls are red in the sample, then 1/3 of balls are red in the population from which the sample was taken).

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Generalization: Inductive inference from a sample to a general conclusion (if all the observed birds do nest all the birds are considered nestmakers).

Projection: Inductive inference from past samples to future samples (If the past three trains have been late, the next one will be late). Distinguish from prediction: projection is one way to do predictions.

Modus ponens: Deductive inference of the form: (i) if A then B, (ii) A, therefore (iii) B. (E.g: (i) If he bought ice cream, then he bought strawberries, (ii) he buys ice cream, therefore (iii) he bought strawberries).

Modus tollens: Deductive inference of the form: (i) If A then B, (ii) not B, therefore (iii) not A. (E.g: (i) If it rains tomorrow the festival will be cancelled, (ii) The festival was not cancelled, therefore (iii) It did not rain).

Hume's problem of induction (1748):

1. Every inference is either an induction or a deduction.
2. To justify an inductive inference rule "I", this rule itself has to be inferred from some premises.
3. The rule "I" cannot be inferred deductively, because there is no necessary connection between past and future premises.
4. Thus, rule "I" must be inferred inductively.
5. When inferring the rule "I" inductively, we must appeal to another (inductive) inference rule "J" to justify this induction. But to justify "J" we need another (inductive) inference rule "K", which leads to an infinite regress.
6. CONCLUSION: No inductive inference rule can be justified.

Infinite regress: A never ending chain of propositions being justified by other propositions which in turn are justified by other propositions and so on.

Foundationalism: Propositions are justified by being inferred from foundational premises which do not need additional justification, for instance, necessarily true premises. Hume's problem has a huge impact in this view of justification.

Coherentism: Propositions are justified by being compatible with a coherent set of propositions, were each proposition in the set is compatible with every other proposition in the set. This view of justification allows to sidestep Hume's problem without denying it.

The hypothetico-deductive method: It is a inductive method since includes both deductive and inductive steps and inductive attempt against truth preservation.

1. Formulate a hypothesis H.
2. Deduce observable consequences {Ci} from H.
3. Test whether {Ci} is true or not.
4. If {Ci} is false, infer that H is false.
5. If {Ci} is true, increase confidence in H.

Falsification: Rejecting a hypothesis as a result of an empirical test.

Confirmation: Increasing the confidence in a hypothesis as a result of an empirical test. Confirmation comes in degrees. These degrees can be measured qualitatively or quantitatively, for instance with probabilities. However according to frequentism hypothesis are the wrong kind of thing to be said to have probabilities since for them probabilities are frequencies of repeatable observable events.

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Hypothesis: A proposition that is either true or false, is not a tautology (not necessarily true – or false –) and that either have some generality and/or is about something not directly observable.

Direct observation: Observation of objects and properties which are accessible through the use of human senses.

Relevant consequences for a hypothesis: they allow us to either confirm or falsify an hypothesis. An irrelevant consequence are those that we expect to be true regardless the truth or falsity of the hypothesis. To falsify a hypothesis one must check all relevant hypothesis.

Operationalization: A way to measure something that cannot be directly observed or that cannot be observed directly with enough precision, by connecting this feature with something that be observed directly.

Asymmetry between falsification and confirmation: No amount of confirming observations can deductively confirm a hypothesis or theory, but one falsifying observation can deductively falsify it. It is not possible to infer the truth even if there are a huge amount of true consequences but a single consequence is enough to falsify.

Corroboration: A hypothesis is corroborated if it has withstood multiple falsification attempts.

Falsifiable: Property of a hypothesis or theory that implies that it might be false, even if it has not been tested yet. A hypothesis or theory is falsifiable if it has implications whose truth or falsity can be determined by means of observation.

Falsification: The event of observing that an implication for a hypothesis is not true and inferring from this fact that the hypothesis is not true.

Falsificationism: The view that science should proceed only through valid falsification, and never use confirmation.

Karl Popper's falsificationism: According to Popper, a theory that is not falsifiable is not a scientific theory. The more falsifiable a hypothesis is, the better it is. In Popper's view confirmation plays no role, he argues that neither the truth can be inferred by seen its implications nor one's confidence in the truth of that hypothesis can be increased. This allows to avoid Hume's problem for induction since only deductive methods are used: implications and testable consequences are obtained by modus ponens and falsification from modus tollens. Main problem is that Popper's deductive program does not allow to scientist to distinguish between non-falsified hypothesis based on their level of confidence, all of them are just non-falsified hypothesis, all kind of judgement or competition between which hypothesis is more likely to be true than other are suspended. In order to face this problem Popper's introduced "measure of corroboration" that distinguishes between non-falsified hypothesis regarding how many unsuccessful attempts of falsification the hypothesis has gone through. Main constrain to this is that it is unclear how could this fit in a pure deductively program. The second worry concerning Popper's deductive program is the Duhem/Quine thesis.

Demarcation of science: Distinguishing science from non-science by providing criteria for counting something as science.

The Duhem/Quine thesis: It defends that we never test a hypothesis alone, but a conjunction of the main hypothesis and several auxiliary hypotheses. Thus, the deduction of observable consequences from the main hypothesis is only possible in conjunction with a set of auxiliary hypotheses. Therefore, only the conjunction which complies both the main hypothesis and the auxiliary hypotheses can be falsified by an observable consequence via modus tollens. This implies that either the main hypothesis or one (or

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several) auxiliary hypotheses are false or even both are. This introduces judgement of confidence in the falsification step of the HD method if the main hypothesis want to be falsified which breaks down the asymmetry between falsification and confirmation. Hence, this is an argument against Popper's suggestion.

Auxiliary hypothesis: A hypothesis used to test another hypothesis, but which one does not intend to test, for instance background assumptions necessary to infer the empirical conclusion.

Conjunction: Two propositions joined by the logical operator AND. The conjunction is true if and only if both propositions are true.

Ad hoc modification: A modification of a claim/hypothesis whose only purpose is to prevent its falsification. It reduces the claim falsifiability. The modification of a claim is ad hoc if (i) the claim has previously been falsified, (ii) the modification saves the claim from this falsification and (iii) it makes the claim less falsifiable – i.e. it does not allow deriving any new testable consequences. Ad hoc modifications must be avoided in science.

Under-determination: An inference is underdetermined if multiple conclusions would be supported by the premises. In order to avoid that severe test is introduced (e.g: Stomach ache is due to stomach cancer).

Severe test (Deborah Mayo): A test in which the probability of observing the consequence would be very low if the hypothesis were false, i.e. C confirms H because C would have been very unlikely if H had been false. That is, the possibility of observing C would have been remote if H were false and some other hypothesis true.

Base-rate fallacy: Mistaken judgments of confirmation based on initial confidence for the hypothesis to be true. This is faced with the Bayesian approach. Initial confidence in an hypothesis is not taken into account when performing a statistical hypothesis test.

3. OBSERVATION AND MEASUREMENT

Empiricism: Sensory experience is considered to be the ultimate basis for knowledge.

Logical empiricism: Theories are created by inferences from fundamental observational statements.

Direct observation: Sense experience of the event itself (e.g: touching a liquid to measure its temperature). It usually only measure qualitative properties.

Aided observation: Sense experience of the event itself with the help of a tool (e.g: microscope).

Indirect observation: Sense experience of the effect of an event, but not the event itself (e.g: bubbles caused by subatomic particles in a fluid).

Theory dependence: An observation is theory dependent if it requires theoretical assumptions in order to make and justify observations. At least some observations are theory dependent. All of the indirect observations are theory dependent since we cannot justify that we observed something indirectly without also assuming that the underlying theory, i.e. the connection between the observable consequence and the property we wanted to measure, is true. At least in some cases aided observation are also theory dependent as some knowledge is needed to understand how the aid object works (e.g: optics theories are assumed to be true in order to observe through a telescope). This introduces a problem when saying that all theories are justified through observations since observations also need underlying theories so

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instead of having one built on the other we have a circularity relationship between the two (foundation theories cannot be found anymore). These cases can be described as nomic measurements.

> **Nomic measurements:** We want to measure quantity X. Quantity X is not directly observable so we infer it from another quantity Y, which is directly observable. In order to do so we need to rely in a relation between X and Y, it is that X can be expressed as a function of Y. The form of this function cannot be derived empirically since both values of Y and X would need to be known and X is the property we want to measure so it is unknown. It can be said that this is a general formulation of the circularity problem. One example on how to deal with it is by using coherentism (e.g: temperature scale).

Operationalization: To operazionalize a property of interest is to provide a way of linking it to a directly observable effect through a correlational relationship or a hypothesized causal chain. Quality critera for operationalization are: (i) property of interest need to be properly defined (independent to the way of measuring), (ii) a valid relationship between the property of interest and the effect which is measured is needed, (iii) there cannot be too many disturbing factors, it is the relationship between property and effect need to be stable when changing background conditions and (iv) a property which is directly observable with enough precision need to be chosen to operationalize. Indirect measures require operationalization and even for directly observable properties scientist often prefer to operationalize them.

> **Property of interest:** The property that is to be measured.
>
> **Hypothesized causal chain:** The supposed link between the property of interest and what can be directly observed.

Operationalism: A view that all basic properties of interest are defined through the operation by which they are measured. The definition of the property of interest is, thus, not given independently to its operationalization. Advantage: it avoids theory dependence problem of measurement since no justification for the relationship between the property definition and the way to measure it (directly observable effect) is needed anymore since it is the definition itself. Disadvantages: (i) it multiply concepts since the same property can be defined as two different things if two ways of measurement are used (e.g: length measured with a ruler or in the space), (ii) it is not possible to critically evaluate operationalizations since if a different operationalization is used (a different way of measuring it) it is considered to be a different property and thus, results can neither be compared nor expected to be the same. If a non-operationalist view is used two different operationalization ways can be used to measure the same property and, thus the results can be critically compared and are expected to be the same no matter the way of measuring it. This is the so-called measurement robustness.

Measurement robustness: Independent ways of measuring the same concept yield the same results.

Measurement: It expresses the result of an observation quantitatively. It complies four steps: (1) definition of the concept to be measured, (2) specification of an operation that is linked to a directly observable property, (3) specifying a measure and (4) defining the measure's units, it is determining what type of scale is used. A measurement has to be comparable to what is being observed and stable over time.

> **Comparability:** A measure has to be comparable to what is being observed.
>
> **Stability:** A unit remains unchanged over time.

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Ordinal scale: Objects are ordered according to a qualitative comparison with regard to a particular property. Ex: Mohs scale of mineral hardness (8 is harder than 6 and 6 is harder than 6 but 6 is not as far from 8 than from 4). It can be transformed through positive monotone (change range).

Interval scale: Objects are ordered with regard to a particular property but the distance between them is also defined. Ex: Celsius or Fahrenheit (0ºC is not the absence of temperature and 4ºC is not twice as hot as 2ºC, but distance between 6ºC and 4ºC is the same than between 2ºC and 4ºC). It can be transformed by using positive linear transformations (multiplying for >0 and adding a positive or negative number).

Ratio scale: Objects are ordered, distance between them is quantified and has a zero point which represents the absence of the property. Ex: Kelvin (0K represent the absence of kinetic energy and 4K is twice as hot as 2K). They can be transformed by using positive scalar transformations (multiplying for a >0 number).

Nominal scale: Classifies objects without ordering them. Ex: categorizing data such as labelling samples.

Absolute scale: A scale where the units for the upper and/or lower bounds are natural. Ex: probability, counting amount of people.

Measurement error: The difference between a measured value and the true value of the quantity. The true value cannot be known in reality.

> **Systematic error:** Error is caused by specific factors that confound the measure process. It can be corrected once it has been detected. The confounding factor can be active depending on the background conditions. It is also called bias or inaccuracy. If there is no systematic error the measurement is accurate. They can be identified by changing background conditions or by comparing different ways of measuring (convergent and divergent validity applies here).
>
> > **Convergent validity:** It is achieved if several, differently operationalized in casually independent ways, measurement processes, all applied to the same target under the same conditions, give the same results.
> >
> > **Divergent validity:** It is achieved if an operationalization yields different results when measuring substantially different targets.
>
> **Random error:** Error is caused due to fluctuations of the measurements for reasons that cannot be controlled or that will not be controlled due to practical reasons. They will all individually deviate but if enough measurements are done the true value can be obtained. If a measurement does not suffer from random error it is precise. It can be avoided by repeating measurements many times.

4. EXPERIMENT

Observation: Registering an empirical phenomenon.

Manipulation: Changing a variable in a study.

Intervention: Changing an independent variable.

Control: Reducing the influence of disturbing factors/variables from influencing the outcome.

Independent variables: Variables that are hypothesized to affect the dependent variable.

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Dependent variables: The variable that changes depending on the changes to the independent variable.

Experiment: Type of observational process characterized by three features: (1) Control of background variables achieved through manipulation, (2) Intervention on the real target (independent) variable identified in the hypotheses that are being tested, also achieved through manipulation and (3) Observation of differences in the dependent variable produced by the intervention. The difference between an experiment and an observational study is the manipulation for the intervention, which observational studies lack. Formal definition: "An experiment is a controlled observation in which the observer manipulates the real variables that are believed to influence the outcome, both for the purpose of intervention and for the purpose of control".

Method of difference (John Stuart Mill): Creating two identical groups and making an intervention in only one of them, being this intervention the only different between the two group, in order to study the causal effect of the intervention. Following step can be followed:

1. A phenomenon E is observed and its cause want to be known.
2. A hypothesis is created by making an informed guess about C being the cause of E.
3. Two identical scenarios S1 (treatment group) and S2 (control group) are created, none of them containing C nor E. All identifiable potential causes for E are controlled in these scenarios.
4. Intervention on C in S1 but not in S2.
5. Observing whether this intervention produces any change in the outcome of S1 with respect to S2, specifically if E is now present on S1 and not in S2.
6. If E is now present in S1 and not in S2, it can be inferred that something caused E in S1, whereas this cause was not operating in S2.

If no control group was used it could have been any other factor that caused E and not the intervention, since this intervention in C is the only difference (assuming all relevant factors are under control) between the two groups, it can be inferred that this intervention in C was the cause for E. Possible errors are: (i) Assuming that all the relevant background factors are controlled even though actually they are not, (ii) Failing to make the intervention, (iii) Failing to make correct observations of the outcome.

> **Control group:** A contrast class that is not subjected to the intervention, but is similar to the treatment group in all other relevant aspects.
>
> **Treatment group:** A class that is subjected to the intervention.

Observational study: An observational process characterized by control of the background factors and observation of an outcome, lacking an intervention. They often allow lower degree of control (e.g: social studies f.i. wages and load of work).

> **Natural experiment:** An observational study in which circumstances are as if there had been manipulation for intervention and control but no manipulation is in fact done (e.g: diseases in water wells).

Field experiment: An experiment where there is manipulation for intervention, but where several background variables are not controlled through manipulation (e.g: quality of soil). Background variables are often taken as given and the only thing that is being manipulated is the intervention.

Laboratory experiment: An experiment where there is manipulation for intervention and where all or most relevant variables are controlled.

Simulation / simulation experiment: A model study (generally programmed in a computer) where the model is created as to mimic an experiment through a representation of the real world, but that it is

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actually a representation and, thus, the variable that is intervened on is only represented, rather than actually present. This last feature is what distinguish a simulation from a laboratory experiment: it is not in the real world but in a representation of it.

Experiment failures: An experiment is considered a failure if we cannot conclude anything of interest through the observation so that there is no evidence that justifies either acceptance or rejection of the hypothesis. It is NOT considered a failed experiment any experiment that does not contain any relevant errors in the experimental design and shows that the expected phenomenon does not occur, as it allows to reject the hypothesis.

Scientific study: An experiment, an observational study or a model.

Internal validity: Property of an scientific study in which the relation between the studied factor and the observed effect inferred from the study is true and it is not confounded by uncontrolled background factors. It refers to the validity of the inferences that are drawn from the study and its assumptions to the conclusion. Typical errors concerning internal validity are failing in controlling and/or identifying background variables, failing to do the intervention as desired or failing to correctly observe the outcome. It is affected by the quality of operationalization. Some examples of errors are confirmation bias, observer effect, placebo effect and selection bias. These can be detected through repetition, reproduction and replication.

> **Observer effect / Influence problem:** The act of observation changes what is being observed (e.g: thermometer reduces the temperature of the liquid we want to measure temperature from).
>
> **Confirmation bias / Interpretation problem:** The observation is registered incorrectly due to psychological properties of the observer.
>
> **Placebo effect:** The therapeutic effect that an inert substance or treatment, designed to have no therapeutic value, has on a patient. It arises when patients' beliefs about the effectiveness of the treatment intervention make a difference in the result.
>
> **Selection bias:** A factor influences the selection of the sample, or the division into test and control group, which one did not aim to create.
>
> **Repetition:** Performing a study again, exactly following the description of the original study. It is only possible if enough information about the setting, intervention, control and instruments used is available.
>
> **Reproduction:** Repeating a study and obtaining the same result as the original's.
>
> **Replication:** Performing a variation of a study with the intention to achieve the same result, where some aspect of the study has been changed to learn how this aspect influenced the result.

Some ways of experimental control are

> **Constancy:** Holding background factor constant between test and control group or between trials in a study.
>
> **Elimination:** Removing the influence of a background factor (e.g: vacuum to remove influence of gravity)
>
> **Blinding:** Eliminating the observer effect through limiting information about the study to participants and/or observers.

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Single blinding / Subject blinding: Blinding the participants in a study, f.i, about whether they are in the test or control group.

Experimenter blinding: Blinding the observers in a study, f.i. about which subjects are in the test and control group.

Double blinding: Both subject blinding and experimenter blinding are implemented.

Separation: Registering the isolated effect of a background variable.

Random sampling: Using a random process to collect a sample from the population. Note: It is different than randomization.

Randomization: Using a random process to divide between treatment and control group. It makes blinding easier, it also helps to eliminate selection bias and to convince other that the test/control group division was not rigged. Although it is helpful, randomization is not necessary for any of these objectives. Experiments that are set by this way are called Randomized Controlled Trials (RCTs) and are considered in many cases as the highest quality evidence available in the evidential hierarchies. However, it is important to notice that (i) randomization does not guarantee the control of unknown factors and that (ii) it does not necessarily include random sampling so the results of a RCT might not be valid for the whole population.

Evidential hierarchies: A ranking of types of evidence based on the processes that produced this evidence.

Data stratification: Dividing a sample into categories based on specific properties before randomizing, thus ensuring that this properties are correctly represented in the test and control groups, f.i. mirroring their distribution in the population.

Known factor: A factor which experiment is aware of. It is a relevant background factor.

Unknown factor: A factor which is a relevant background factor but the experimenter is not aware of.

External validity: Property of a scientific study by which the conclusion obtained holds not only for the system but also for the target, f.i. an inference from a sample to a population. It is related to the inferences that are done from a specific study to the target or the population of interest. Typical check points are whether the intervention in the study is similar to the intervention in the real case or if there is an important difference in the background variables between study and real case.

5. MODELLING

Model: It is a representation of something from the real world, a target, but that is different from the actual target concerning one or several properties, i.e. they are idealizations. The nature of these differences depend on which the model is going to be used for, i.e. models are purpose dependent. It can be a material model such as scale model (different size) or animals that are used as models, simulations or computational models and mathematical models. Reasons for using models can be several such as the real target cannot be handled directly due physical or economic reasons, it is morally or legally restricted or prohibited to use the real target (e.g: humans) or the actual target is too complex to perform the required study. Unlike for experiments, in modelling the most troubling affect to external validity i.e. whether the relation between model and target holds or not.

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Representation: Something that stands it for another thing (the thing that it represents) in order to be used to perform an investigation or a study in that thing instead of in what we are really interested in.

Target: The thing that a model aims to represent.

Material models: Made of real existing matter.

Idealization: The model and the target differ in some of their properties, it does not have all the properties the target has and vice versa. Description of these differing properties are called idealizing assumptions.

Purpose dependence: A model is justified regarding how useful it is for fulfilling a certain purpose. This means that the quality of a model does not depend only in the model-target relationship but also on the model's purpose, hence there can be different good models for the same target if their purposes are different. Models are purpose dependent, theories are not.

Analogical inference: (i) Establishing that a model and a target are similar in certain properties (positive analogies) and dissimilar in others (negative analogies), (ii) Identifying other properties {Q1,…,Qn} in which it is unknown whether the target has them or not (neutral analogies), (iii) Drawing the conclusion that the target also has the property Qi because it arises from the positive analogies and it is not disturbed by the negative analogies. Thus the three requirements can be summarized as: (1) There are positive analogies between model properties and target properties, (2) The neutral analogy Qi that is transferred from the model to the target must arises from the model properties being positive analogies and (3) The causal relation between the model properties and Qi, including all preconditions, must not be part of the negative analogies. Model manipulation is used to discover neutral analogies.

Analogy: Inference by analogy is to state that a conclusion in one case applies to another case due to there being relevant similarities between the cases.

Positive analogy: Similarity between model and target.

Negative analogy: Dissimilarity between model and target.

Neutral analogy: Property of model, whose occurrence in target is not known.

Epistemic virtue of models: Types of properties of models that determine the quality of a model given a specific purpose. There exists trade-off between them. Most important ones are: similarity, robustness, parameter precision, simplicity, tractability and transparency.

> **Similarity:** The degree to which the model has several properties that the target also has, i.e., there are multiple positive analogies, that are relevant for the modelling purpose. The more similar the model is to the actual object that it is supposed to represent, the better the representation is. Only similarity in properties which are relevant for the model purpose matters here.
>
> **Robustness:** A model is robust with respect to an assumption if changing this assumption does not change the model result. If a model is robust with respect to an assumption we can be quite confident that the specification of that assumption is irrelevant for our purpose. However that does not mean that we can erase that assumption.
>
> **Parameter precision:** One model has higher parameter precision than another model if the specifications of the parameters of the first model implies the parameter specifications of the second. It is thus a property of the model itself, not of the relation between model and target.
>
> **Simplicity:** One model is simpler than another if it contains less parameters than another model.

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Tractability: A model is tractable with respect to some general set of rules if the relevant model result can be obtained by applying these principles to the model.

Computational tractability: A model is computationally tractable only if its result can be obtained in polynomial time.

Analytical tractability: A model is analytically tractable only if the model result can be obtained through valid deduction, from the model assumptions alone. Mathematical models are an example of this.

Theoretical tractability: A model is theoretically tractable only if the model is computational or analytically tractable, and some of the necessary (computational or deductive) steps are justified with reference to a background theory.

Transparency: A model is epistemically transparent if the model user is cognitively capable of understanding how the model result is produced. It is desirable because it allows checking the correctness of a result.

Mirror models: A model with high similarity to the target. Models that are very similar to the target and very precise but often compromise on simplicity, transparency and tractability.

Isolation models: A model with high simplicity. They represent isolations of particular features of the complex world. It is important to make sure that the studied system is separable in this way and does not suffer from interaction effects.

6. STATISTICS

Lying with statistics: The intentional misapplication of statistical methods. Statistics offers many different methods, but some of them are appropriate for some problems and other are not. Intentionally misapplying a statistical method in order to give fake results is refer as lying with statistics (e.g: mixing mean and median).

Statistical methodology: Justification of the choice of using a particular statistical method.

Reasons for using statistics: (1) Some hypothesis only have stochastic implications, i.e. they do not say anything about the world with certainty but rather only say things that may happen with a certain probability. (2) Sometimes a quantitative measure of measurement error is desired and (3) It helps to express differences in confidence in a hypothesis.

Descriptive statistics: Statistics used when one wants to display data and conclusion accurately.

Inferential statistics: Statistics used when one aims to draw a justified conclusion from the data.

Stochastic hypothesis: A hypothesis whose implications come in the form a probability distribution.

Deterministic hypothesis: A hypothesis all of whose implications are certain. They imply that something is the case or not with certainty.

Quantitative measurement of error: The likelihood of a measurement error being made, presented on a quantitative scale.

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Error based statistics: Determining the probability of an observation given that a certain hypothesis is true.

Confidence in a hypothesis: The subjective estimation of the probability of a hypothesis.

Fisher's significance level: A method of statistical hypothesis developed by Ronald Fisher. The steps are the following

1. You specify the experiment you want to run.
2. You specify all possible outcomes for that experiment.
3. You categorize all these outcomes by the value of test statistic (f.i.: number of heads in coin flipping).
4. You calculate the probability of each value of the test statistic by summing up the probability of the outcomes that yield to it. This gives you the sampling distribution of the test statistic, under the assumption that the main hypothesis is true.
5. You perform the experiment and observe the result.
6. You sum up the probabilities of the outcomes that are at least as extreme as the ones you observe (p-value).
7. If the p-value is smaller than a imposed threshold (significance level), you reject the claim.

This test can be manipulated by (i) testing a null hypothesis and hoping that rejecting it gives some support to yours even though it does not, (ii) redoing the test as long as you operate with a low-powered test, (iii) assuming a different sampling distribution, (iv) setting a different significance level.

Test statistics: A quantity, computed from values in a sample, that is considered for a statistical purpose.

Sampling distribution: A distribution over the possible outcomes of the test statistics.

p-value: The probability of observing an outcome at least as extreme as the observed outcome.

p-value abuse: Changing test set up, statistical method, or sample in order to make the p-value either higher or lower than the significance level, depending on which result is desired.

Significance level: A conventionally set level for p-values below which the associated hypothesis should be rejected.

Neyman Pearson hypothesis testing /approach: It compares two hypothesis that are mutually exclusive and jointly exhaustive. In principle, either of these hypotheses might be true and accepting one implies rejecting the other. Thus there are four possible outcomes:

a) You correctly accept the true hypothesis.
b) You correctly reject the false hypothesis.
c) You wrongly reject the true hypothesis (Type I error).
d) You wrongly accept the false hypothesis (Type II error).

Before performing the test you can choose the probabilities of making each kind of error. However there is a trade-off between them since type II error rate depends on type I error rate. This trade-off can be weakened by increasing the size of the effect that is being tested and by increasing the sample.

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This is formally measured by the so called power of the test which is the probability of correctly rejecting a false hypothesis.

Differences between Fisher significance test and Neyman Pearson's procedure: The probabilities choice for type I error and type II error are transparent and can be justified whereas p-value abuse in the Fisher test is not. Fisher significance test only tests one hypothesis and its decision criteria is arbitrarily chosen and, hence, open to manipulation whereas NP is not.

Original hypothesis (Hi): Some claim that you are interested in.

Alternative hypothesis (Ha): A hypothesis that due to logical necessity has to be true if the original hypothesis is false and vice versa. It is the inverse or negation of Hi.

Type I error: Wrongly rejecting a true hypothesis Hi.

Type II error: Wrongly accepting a false hypothesis Hi.

Power of a test: The probability of correctly rejecting a false hypothesis Hi. It is the inverse of the type error II rate.

Bayesian statistics: Posterior probability of a hypothesis is calculated based on the prior probabilities together with the observed outcome, using Bayes' theorem. Steps are as follows:

1. You formulate the hypotheses. In the Bayesian framework there is no limitation in the amount of hypotheses one can test, but all of them should be mutually exclusive, i.e. only one can be true at the same time.
2. You determine the so-called prior probabilities p(Hi) for each hypothesis.
3. You decide which experiment you want to run.
4. Considering the experiment, you determine the values of test statistic, the possible outcomes and the probabilities of the values if Hi were true.
5. You run the experiment and collect the data.
6. You calculate posterior probabilities to the hypotheses. The idea is that observation should affect your beliefs but gradually, i.e. the more observations you make the more your subjective beliefs take into account how the actual world is.
7. The hypotheses are not rejected nor accepted, probabilities with which they are true or false are assigned to them.

Problems with Bayesian statistics are the following

The problem of priors: Bayeasianism does not offer a clear way to determine prior probabilities. The principle principle and the principle of indifference apply here.

The principle principle: A subject's prior probability should be assigned on the basis of objective probability, if it is known.

The principle of indifference: A subject's prior probability should be assigned equally to the possible outcomes, if there is no information about the objective probability.

The problem of slow convergence: If two subjects assign sufficiently different prior probabilities to the same hypothesis, it is possible that their respective posterior probabilities will not converge even though Bayes' theorem has been applied to large amount of data.

The problem of old evidence: The problem of determining what evidence has been previously used to determine posterior probabilities. That evidence should not be used again.

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The problem of uncertain evidence: Bayesianism does not take uncertainty about evidence into account, for instance, failing to correctly read an observable effect.

Different ways of expressing the same risk:

A	B	C
Relative risk reduction	Absolute risk reduction	Number needed to treat-format
"Our medical intervention results in a 33 % relative decrease in the incidence of fatal and nonfatal myocardial infarction."	"Our medical intervention results in a 1,3 %-point decrease in the incidence of fatal and non-fatal myocardial infarction – from 3,9 % to 2,6%."	"We must treat 77 persons to prevent 1 fatal or nonfatal myocardial infarction."

7. EXPLANATION AND CAUSES

Explanation: It is a mean to generating understanding. It constitutes one of the central aims in science.

Explanandum (explananda): That which is to be explained.

Singular explanadum: A singular event, phenomenon or property that is to be explained.

General explanandum: A general event, phenomenon or property that is to be explained.

Explanans (explanantia): That which provides the explanation.

Singular explanans: The explanation is provided by a number of particular events, phenomena and properties.

General explanans: The explanation is provided by a regularity or a law.

Deductive-nomological (DN) account of explanation: According to this account an explanation is a deductive argument derived from natural laws or law-like generalizations combined with initial conditions from which we deduce the phenomenon to be explained. The name comes from (1) laws of nature play a central role (nomological) and (2) it relies on making deductions (deductive). The main difference between explanations and predictions according to this account is the timeline: predictions are about the future, i.e. giving reasons for expecting a phenomenon to happen, and explanations are about the past, i.e. giving reasons why we should have expected the phenomenon to happen.

Natural law: A scientific law, usually thought to capture fundamental relationship in the natural world.

Law-like generalization: A generalization based on empirical evidence which appears to capture regularities in a way similar to how a law does.

Problems with DN account of explanation: (i) It does not take into account asymmetry in the productive relationship between two phenomena, (ii) It does not separate casually relevant from casually irrelevant factors (e.g: a man trying to get pregnant and taking birth control pills), (iii) It can identifies deductive arguments as explanations although these do not produce understanding, (iv) an explanation does not need to fulfill DN requirements since a simple sequence of events can sometimes provide useful explanation and understanding without any regularity/law.

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Casual account of explanation: It argues that the important thing of generating understanding is to be able to answer the question "what if things would have been different?". According to this account, causes are the main component of the explanans, specifically causes that make a difference with respect to the phenomenon that is to be explained. Symmetry in equations is misleading, we can obtain one parameter from another symmetrically but that does not mean that one provides an explanation for the other symmetrically, it is unidirectional (e.g: pendulum, period and length). Thus, understanding an explanandum is not achieved by laws of nature but by showing how it was produced and by answering how this productive relationship would have change under counterfactual conditions. The main point is to provide a contrastive explanandum and to identify the difference-making casual factors, this is what is needed for generating understanding.

> **Explanation should be contrastive.** This contrast should be made explicit so that the explanation can directly answer to the relevant difference-making factors.

Explanatory virtue: A property of an explanation which improves its quality in some respect. It makes the explanation powerful.

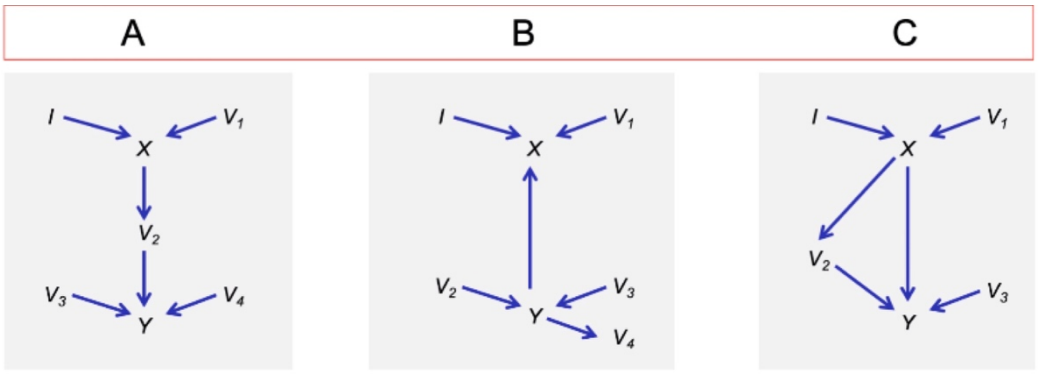
> **Accuracy (explanations):** The explanans is true in the relevant aspects. It only concerns the difference – making factors, not all features.
>
> **Precison:** This criterion is concerned both to the explanandum and to the explanans. They must pick out the difference – making causes.
>
> > **Precision of the explanandum:** The contrast featured in the explanandum is precisely stated.
> >
> > **Precision of the explanans:** The contrast featured in the explanandum is reflected in the explanans. It should state the difference – making causes and avoid including conditions that did not make a difference in the phenomenon to be explained.
>
> **Non-sensitivity:** The causal chain in the explanans is sufficiently short so that the explanans is not sensitive to small changes in circumstances. Causes which are sensitive to long and intricate chains of events should be avoided.
>
> **Cognitive salience:** The explanation is easy to understand for a target subject.

Causal explanation: A productive relationship between two events that is asymmetric and thus different from a symmetric mathematical equation.

Direct cause: X is a direct cause of Y if and only if there is a possible intervention on X that will change Y, when all background conditions are kept the same (image C, in image B Y is a direct cause of X).

Contributing cause: X is a contributing cause of Y with respect to some background variables if there is a casual chain such that each link in the chain is a direct cause between two variables and where the chain begins at X and ends at Y (image A). Usually they have more explanatory value.

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Causal explanations normally refer to contributing causes rather than direct causes. However the longer the cause chain is, the more sensitive the explanation becomes (trade – off).

Correlation vs causation: Correlation is a measure of association or co-variation between two variables, whereas causation is a productive relationship. Correlation is symmetrical whereas causation is asymmetrical. Correlation is neither necessary nor sufficient for causation. However, causation is difficult to observe as such, while correlation is an important form of evidence for inferring a causal relation, even though any given correlation is theoretically compatible with multiple causal scenarios.

Justification for causal scenarios: Any given correlation is compatible with multiple scenarios, hence it is necessary to justify causal relationships. For that Mill's method of difference can be used, also observational studies are valid. Instrumental variable analysis is often used.

> **Instrumental variable analysis:** Determining causal relationship from data. You observe a correlation between X and Y. Then you identify another variable Z that produces a change in X but does not have a direct causal influence on Y. Now you use the correlation between Z and Y in order to determine if X causes Y or not. An experiment is not necessary, observational studies are also valid for this mean. We need causal knowledge in order to produce more causal knowledge.

8. ENGINEERING DESIGN

Difference between sciences and engineering: Sciences aim to understand and explain the world and they use abstract reasoning and theory from the start. On the contrary, engineering aims to create artefacts, its function is not to explain the world but to change it. While science (previously considered as natural philosophy) have always been taught through lectures and books, engineering required, traditionally, specific and local knowledge that was acquired through trial and error, i.e. tacit knowledge was a requirement for engineers. However, in modern engineering this knowledge is systematically developed, it is there is a burgeoning engineering theory.

> **Tacit knowledge:** It is a kind of knowledge that comprises skills, ideas and experiences that people have but which are not codified and thus are difficult to transfer to another person by verbalizing it.

Artefact: Existing material object that has been assigned one or multiple functions. They can be both natural or human made, same as non-artefacts (e.g: stone used as a hammer is an artefact and non-natural chemical elements are non-artefact but human made). The ascribed function is thus a necessary

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characteristic. Moreover, they are material objects and therefore they operate through cause-effect relations that are governed by laws of nature.

Technical artefact: An artefact which realizes its function in virtue of its physical properties. They have a dual nature as they are characterized both by their physical properties and by their intended functions.

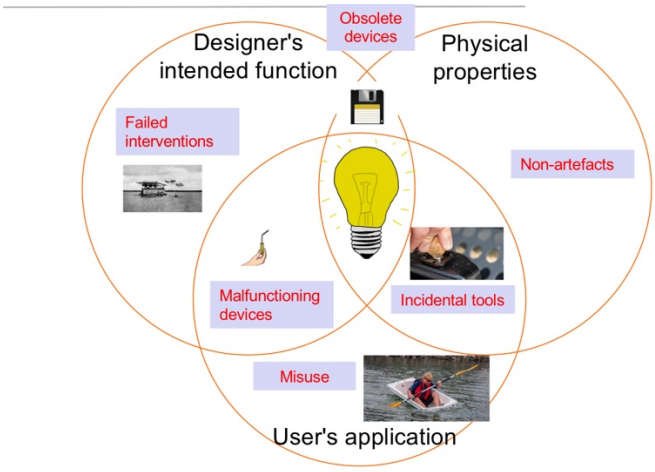
Dual nature: Technical artefacts are characterized both by their physical properties and by their intended functions.

There are some problems regarding this definition, namely the ascribed functions can be given by both the designer and the user. Different cases can be seen in the figure, where all the circles collapse is the standard definition of (technical) artefact. Thus the possibilities for artefacts are: malfunctioning devices, incidental tools and obsolete devices.

Malfunctioning artefact: An artefact that has an assignment such that it ought to satisfy certain functions of a given functional kind but which have lost the physical properties that supported this function. It has no longer the needed physical properties in order to satisfy the function that was assigned to it.

Misuse of an artefact: An user employs an artefact in a way that does not realize the function of its functional kind, i.e. the functions assigned to it.

Functional kind: A categorization of artefacts and their descriptions in terms of the functions they ought to satisfy.



Function: The purpose that someone intends for an artefact. They are established by people's intentions and can be given by both user and designer.

Normative: How things should be, regardless of how they are.

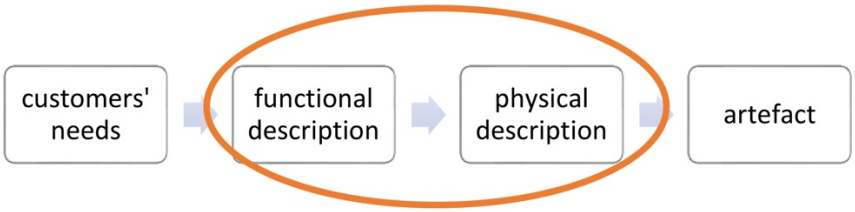
Ascribed functions: A function given to an existing object by an USER who believes that this object has certain capacities that can fulfill a certain purpose. Ascribed functions have a weaker normative claim.

Function assignment: A function given to an object by a DESIGNER by creating or describing an object with certain capacities that can fulfill a certain purpose. Assigned functions have a stronger normative claim.

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Design: A process that develops a physical description corresponding to a functional description. It generates an artefact by bridging the functional and physical description that corresponds to a functional description. The design refers to a particular knowledge that often needs to be generated and this is also part of the design process.

> **Creation – centered:** A design process creates a physical object that robustly satisfies a functional description.
>
> **Client – oriented:** A design process creates a physical object that robustly satisfies the client's needs.



First arrow: What functions should the device satisfy? You need to develop the functional concepts, understand those dimension of functionality and learn what do your clients want from each those dimensions.

Second arrow: Which physical structures might satisfy the demanded function? Operational principle applies here. Engineers also engage with device-restricted simplifying theory rather than pure scientific theories.

> **Operational principle:** The underlying, basic concept of why a particular physical structure might deliver a certain functionality.

Third arrow: The production knowledge. A real separation between science– and engineering–knowledge is found here.

> **Difference between science knowledge and engineering knowledge**:

ENGINEERING KNOWLEDGE	SCIENCE KNOWLEDGE
Investigate functional demands	No concept of technical function
Work with concrete functional objectives	Few rigidly specified goals beyond understanding
Organize theory and data around functional kinds	Organize theory and data around natural kinds
Allows for device- or context-specific knowledge, where convenient	Always aim for generality
If the devices to which they apply cease to be useful, the knowledge will no longer be of interest	Once acquired knowledge always of interest

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Design methodology: Method choice within a design process.

1. What functional descriptions?
   An appropriate functional requirement has four conditions: (i) It should be a functional description + implementation independent, i.e. it should describe what it is for but without already giving a physical restriction on its physical implementation. (ii) It should be complete, i.e. all the desired effects of the artefact should be properly described. (iii) It must take care of the use – context, i.e. the functional description must depends on the users' behavior and on the way in which users interact with the device. (iv) It should have quantitative criteria by specifying threshold of failure and degrees of success so that designer can refer to these criteria later on in the validation step.

2. How to generate and weed out possible physical descriptions that satisfy functional requirements?
   Physical descriptions are generated by functional decomposition. It starts with a high – level description of a function and then it is divided in several sub-functions. Each of these sub-functions are also decomposed reiteratively. At some point we can have some sub-sub-…-functions for which there exits some standard solution that has already been implemented before and we do not have to worry anymore for those. However we will find some for which we need to implement something new, we would probably have some theoretical or empirical knowledge that we need to use to reject what does not work and to establish some constrain limits when implementing the variation (blind variation). We should not only find one possible answer but several and later on weed them out by using either theoretical or material modelling.

3. How to select the most promising possibilities?
   Here a possibility is to proceed as in an optimization problem and choose the one that performs optimally with respect to the relevant functional requirement. In order to do so we evaluate a large amount of possibilities and then determine which one fits best. However in a practical way this is not actually done, instead the satisficing procedure is used which is more effective than full optimization.

   > **Satisficing:** A process where search and evaluation is combined, so that options that fulfill a certain threshold are chosen. It consists in defining a minimal threshold (can be the one we defined in step 1) and then starting to search possibilities and as soon as we come up with one, we evaluate it with respect to the defined threshold and if it fulfills it we end the search and choose that possibility. Otherwise we continue searching and proceeding the same way.

4. How to validate a design?
   In order to do so we can perform testing.

   > **Design validation:** Testing aimed at ensuring that a product or system fulfills the specified functional requirements under specified operating conditions.

Engineering design: It requires its own knowledge and cannot be categorized simply as applied science.

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9. QUALITATIVE METHODS

Interpretation: It is the central method for the social sciences. It consists in explaining why some agent has a certain behaviour (outcome) given certain instructions (input). Folk psychology explains it through either the simulation theory or the theory theory.

> **Simulation theory of interpretation:** We reuse our cognitive mechanisms (the way we believe, desire and intend) to simulate other people's metal states. We put ourselves in other's shoes.
>
> **Theory theory of interpretation:** We construct a theory of mind just like a scientific theory, using collecting evidence and revising hypothesis.

There are several accounts through which the outcome behaviour can be explained:

> **Belief-desire explanation:** An agent's action A (instead of B) is explained by herself desiring some result X (and not Y) and her believing that A will bring X and B will bring Y. Someone's choice is explained by pointing to her relevant beliefs and desires. Interpretation is considered thus as something else than just an input and an outcome, the inputs are assumed to be mentally processed by the subject and leads to a certain behaviour (output).
>
> **Methodological behaviourism:** The main point is to discover and develop regularities between stimuli and responses (e.g: Pavlolv's dog). Thoughts and feelings, which may or may not exist, are not part of science of behaviour.
>
> **Cognitive science:** It holds that theories refer to mental states, but only if those mental states are properly operationalized with respect to observational variables. Unlike for introspective, for cognitive scientist it is not a problem to attribute a belief or desire to a machine if their behaviour is sufficiently complex that it cannot be otherwise explained. This is the so-called intentional stance.
>
> > **Intentional stance:** Attributing mental states like belies and desires to an agent A (human, animal or machine), assuming A to be rational and using this to predict the actions of A. It consists on figuring out what beliefs and desires that agent ought to have given its space in the world and its purpose and inferring from that what that agent will actually do, i.e. generating a prediction.

Main problem of interpretation: The multiplicity of possible, plausible mental-state assignments for a single observation. They are called alternative mental-state models. This problem can be faced by (1) avoiding interpretation altogether or by (2) improving it. Interpretation is thus a fallible method but it can become more reliable if quality criteria are used.

> **Avoiding interpretation:** Methodological behaviourist can be used when possible (low complexity cases) as it is considered a viable scientific strategy in many domains. Another option is to investigate relation between social properties instead of individual properties (holism) or, lastly, to investigate relation between psychological properties and behaviour, instead of interpreting behaviour itself.
>
> **Improving interpretation:** It can be done by (i) sampling individuals that minimize the need for interpretation or that have a positive attitude toward the study, (ii) looking for more evidences, beyond the instructional input and the behavioural outcome in order to assign a mental state such as implementing thinking outloud protocols, using competent observers, checking for coherence in an interpretation or using intersubjective checks.

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Qualitative data: Data without numerical properties, typically classification or categorization. It has its own important purposes without the need of quantitative data. In order to measure qualitative data it is also necessary to operationalize the measurements by assuming that there is such an appropriate relationship between the mental attitude of the subject and the observable response. This validity can be evaluated through the quality criteria for the qualitative data.

> **Qualitative vs quantitative methods:** The move from the qualitative to the quantitative data is just a refinement of information that is, at least partly, already contained in the qualitative data. Thus, qualitative data can often be quantified and in these cases quantitative and qualitative data share at least part of the informational content.
>
> **Mixed method (of data types):** Using qualitative data both in its own right and as raw material for generating quantitative data of one several properties. One possibility is to focus first in the qualitative data and then proceed to the quantitative which often narrows the focus.

Quality criteria for qualitative data:

> **Ensure repeatability:** It is achieved by ensuring procedure rigor, i.e. to document the techniques of data collections performed and make the form of data analysis used transparent.
>
> **Control observer effect:** It is achieved by remaining as neutral as possible. It is needed since blinding is not possible when doing surveys or interviews as questions are the only available tool
>
> **Control observer bias:** It is achieved through reflexivity. Description of the influence of relation between researcher, topic and subjects as well as researcher's sociocultural position. For instance, the observer can describe the way he thinks he affected the subjects.
>
> **Robustness:** It is achieved through triangulation. It means using multiple methods of observation about the same properties. For instance, combination of revealed and stated preferences can be used in order to aim for convergent validity.

Case study: A study characterized by using a small number of cases that are chosen for some specific reason and display the same outcome. The study typically does not involve experimental manipulations. The goal of the study is to understand a specific case. They can be both qualitative and quantitative studies. Motivations for using case studies are: (i) when operating from a falsification point of view since a single falsifying case is enough to reject a hypothesis; (ii) when the studied phenomenon is sufficiently complex for the researcher not to know how to approach it, in these cases it is used the so called analytical narrative; (iii) when comparing a number of cases without knowing exactly what the underlying population is, in these cases the aim is to know about necessary causes (not sufficient causes) that lead to a better understanding of the boundaries of the population, here the so called qualitative comparative analyses is used; (iv) when studying the causation-effect relationship in an individual case, i.e. what kind of mechanism works in the background when doing an intervention and observing an effect, in order to do so process tracing is used.

> **Analytical narrative:** A kind of case study where a narrative is presented to clarify the important actors and main decision points. It consists on building a model to express the logic of a possible explanation given a phenomenon, typically in the context of a unique case. Specifically the method involves: (1) making explicit research question; (2) selecting cases; (3) using narrative to elucidate the principle players, the inferences, the key decision points and the possibilities; (4) evaluating the model through the testable implications it generates; (5) considering alternative narratives. This method can be improved by considering alternative models for the same case, i.e. alternative narratives.

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Quality comparative analysis: Analysis of cases performed in order to learn about necessary causes and to better understand the boundaries of the underlaying population. The result can be applied in a multivariate analysis, it can be seen as an important step towards the multivariate studies. The focus is on the potential necessary causes of phenomena, in order to do so the cases where the strike is the desired one are compared so that all the compared cases have the same outcome. The main difference with a multivariate study is that in the later the distinction between necessary and sufficient causes cannot be done, rather they are considered as they are already identified.

Process tracing: A kind of case study that emphasizes cause-effect relations in a single case vs cause-effect relations as mean effect size in the population. The steps are: (1) Formulate hypotheses about alternative mechanisms; (2) Work out differences in empirical implications; (3) Check case data for these mechanistic fingerprints; (4) Conclude which mechanism it is.

Single case vs multivariate studies: A representative sampling is more related to multivariate studies whereas an information-oriented sampling is more related to single case studies since they pick out individual cases.

Chosen case: Choosing one individual case among others need to be justified in connection with the background theory and it must be cleared stated why we are choosing that case. It is also important to check beforehand if the case is separate from the context or, in the contrary, it include some of the context.

Transferability: When using a single case there is little (if any) evidence to transfer the result to anything else such as a larger population or another case. This transferability can be improved by identifying if are there other predictions covered by the model that can be tested in that particular case and by performing out of sample tests to check whether the model applies to other cases.

Improve of case studies: It can be achieved by: (i) specify the research question; (ii) specify theory background and hypothesis; (iii) clarify what constitutes a case unit; (iv) make case selection explicit; (v) if there are multiple cases, clarify what makes them comparable; (vi) investigate all the implications of the theoretical hypothesis; (vii) investigate rival theoretical hypotheses.

10. RESEARCH ETHICS

Ethics: It investigates what reason do we have for endorsing certain claims.

Research ethics: It answer to questions related to why is it rational for all scientists/students/teachers/etc to endorse certain codes of behaviour.

Morals vs ethics: Morals are properties of actions, intentions or decisions (being good or bad), whereas ethics is providing justification for why actions, intentions or decisions have this moral property. However, in many usages this distinction is not maintained, for instance the same behaviour might be described as immoral or as unethical, without any clear difference being intended.

Permission: Being morally allowed to do something.

Obligation: Being morally required to do something.

Prohibition: Being morally forbidden to do something.

Descriptive ethics: It studies what people consider to be right and wrong.

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Descriptive view of morality: Morality consists in the code of conduct that are established in a society, a group or among individuals. The problems with the view are (i) the existence of non-moral codes of conduct such as conventions, etiquette, law or methodology; (ii) its inherent relativism since morality as seen from this account depends on people acceptance for any code of conductance and this acceptance can differ between two people.

Normative ethics: It studies what people should do, irrespective of what people actually think.

Normative view of morality: It is based on the universal rational endorsement and take into account counterfactual conditions. It imagines how an individual would decide if his or her situation were substantially different from what it actually is. It assumes/requires that for any event E (i) the individual has perfect information about how E affects any conceivable position a person may occupy in the society or in any other considered group and (ii) the individual does not know in which position he personally is; thus the individual will judge E not based on a personal view. Hence, this view distinguish morality from established codes of conduct.

Universal rational endorsement: Moral code or code of conduct which all rational people, under certain specified conditions, would endorse.

Three frameworks of normative ethics are:

Consequentialism: Choices are to be morally assessed only by their consequences.

Utilitarianism: It is an example of consequentialism which argues that we should assess the consequences of an action by the happiness or desire satisfaction that the action brings about. According to this view, each individual counts the same and has the same worth.

Consequences for science: They include consequences for the epistemic goal of science, for the institution of science and consequences for the uses of the scientific results.

Problems for consequentialism: (i) It is extremely demanding since every action has some consequence, thus for every choice of action, you would have to take into account both consequences for yourself and for others, i.e. consequentialism rests in a very strong claim; (ii) It might intuitively permit wrongful acts. In order to face that last problem consequentialists have developed further sophistications.

Deontology: It says that to choose morally is to fulfill relevant duties. A duty can be to respect rights of others, among others. It does not matter how good or bad some consequences are, certain choices are prohibited, mandated or required in themselves and we are considered to have rational insight into the set of rules and duties that we must adhere to.

The categorical imperative: An action is right if it is in accord with generally defensible principle. It was defended by Inmanuel Kant (s.XVIII) as “If you have a maxim that you think should be a universal law, then you should act according to that maxim”.

Advantages of deontology over consequentialism: (i) It makes life much easier for a scientist who wants to determine whether a particular research project or experimental design is in accordance with ethical rules, he can simply goes through a checklist of rules that are justified by the categorical imperative while consequentialism needs to consider all possible consequences. (ii) There is a finite number of principles, that means that there are certain areas in any person’s life in which choices can be done without raising any moral question.

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Disadvantages of deontology over consequentialism: (i) All rights and duties are categorical, i.e. there is no room for degrees. This makes it more difficult to incorporate risk. (ii) Weighing duties against each other when they conflict is not straightforward, there might be cases where you have to respect a duty but that may violate another duty.

Virtue ethics: Morality consists in exemplifying good, or virtuous, character traits. The idea comes back from Aristotle. A big difference from the two previous views, virtue consists in an insight into human nature, i.e. being a moral person is a way of fulfilling our human nature.

Informed consent: It is a requirement in order to perform an experiment morally. This means that a prospective participant is (i) informed: what the goal of the experiment is, what kind of side effects are known; (ii) participating voluntary (and taking into account social or economic pressures); and (iii) decision-capacitated: it is necessary to make sure that the participants are actually capable of processing the information provided.

Justification for informed consent from each perspective:

Consequentialism	Deontology	Virtue ethics
• individuals know best what is good for them • establish and maintain trust in science	• don't lie • treat others as autonomous beings • be fair to others	• sincerity • reflexivity • respectfulness

Where reflexivity refers to consider your own limitations in the decision and wanting to include others to participate in that decision.

Scientific misconduct: Misbehaviour when performing an experiment or a scientific study. Manipulating data is allowed, all scientists do it in order to clean data and remove outliers, but it should be transparent and well justified so that it is methodologically and morally acceptable. Some examples of misconduct through manipulating data are fabrication, falsification and plagiarism. Acting immorally due to reasons exposed in the table. This can be avoided/minimized through peer review and choosing competent and impartial observers.

Consequentialism	Deontology	Virtue Ethics
• Undermines trust • Wastes resources	Don't lie!	• Sincerity • Humility

Fabrication: Intentionally making up data or results without scientific support to mislead the reader.

Falsification: Intentionally changing data or results without scientific support to mislead the reader.

Plagiarism: The appropriation of another person's ideas, processes, results or words without giving appropriate credit.

Scientific transparence: Freely sharing the material from which the conclusions were drawn.

Whistleblowing: Exposing illegal or unethical behaviour within an organization. The person who exposes it is the so called whistle-blower.

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Authorship: Being an author, or co-author, establishes credit for one's intellectual efforts. It does not only establishes credit, but also responsibility. Putting someone's name in a publication certifies that that publication is the result of that person's work and that that person has performed the work to the best of his knowledge and abilities.

Author: A person is an author of an article if the person has made a substantial contribution to (1) the conception of the work, (2) drafting the work, as well as (3) is involved in the approval of the final version, and finally (4) agrees to be accountable to all aspects of the work, i.e. puts the responsibility for any misconduct on all co-authors.

Ethical justification of authorship:

Consequentialism	Deontology	Virtue Ethics
• Impartiality of science • Public trust • Incentives for scientists • Saving resources	• Intellectual property rights • Responsibility	• Sincerity • Humility • Resoluteness • Reflexivity

Self-plagiarism: Use of one's own previous work in another context without citing that it was used previously. The problem here is that one is presenting something as new while in fact it is not.

Ghost authorship: Ghost authors hide authorship, maybe to avoid responsibility or to conceal conflicts of interest. Reasons why it is not moral according to each view:

Consequentialism	Deontology	Virtue Ethics
• Impartiality of science • Public trust	• Responsibility	• Resoluteness • Reflexivity

Resoluteness: The virtue of staying with one's work gorging on despite of difficulties.

Reflexivity: The virtue of being critical enough of one's work, making due allowances for one's own bias.

Gift authorship: A person who is stated as an author, for instance for social reasons, but who do not fulfil the criteria of being stated as an author. It is also called honorary authorship. The ethical reasons of why this is considered as a misconduct are the same as for the case of plagiarism.

When do we have gift authorship: In general, those who merely run an experiment, provide technical support, collect data, proofread or comment will not qualify as co-author, although they should be mentioned in the acknowledgments, i.e. those who do conceptual contributions can be considered co-authors whereas the rest cannot.

Conceptual contribution: Study design, data analysis decisions and data and results interpretation.

Non-conceptual contribution: Obtaining funds, commenting on drafts, running experiments, provide technical support, collect data, proofreading.

Therefore, for instance, someone who has made substantial contributions to the conception of the work and meets the draft and final approval conditions can be cited as co-author, however some who only obtains the funds cannot.

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Order of naming authors and co-authors: They can be ordered regarding how much substantial contribution have made each of them or, in order to avoid problems, an alternative is to list the exact contributions of each co-author in a footnote of the published paper.

Some cases of ethical **career choices** are related to:

Dual use (technologies): Those goods, products or technologies that are usually used for civilian purposes, but that can be easily repurposed for military purposes, in particular for illegitimate ones.

Ethical justification of dual technologies: Deontologist argue that you must not intend to harm, to deceive or to kill, therefore it is morally prohibited to use/design dual technologies. Vision from virtue ethics is quite similar to the former. In contrast, for consequentialists all the possible consequences need to be considered so it is required to use all the available knowledge in order to consider all those consequences and base the decision on that.

Frivolous research: Research conducted (at least as far as can be determined) purely for the fun of it.

Ethical justification of frivolous research: According to one of the group of consequentialists, the auto-denominated effective altruists, there is a moral obligation to choose those careers and topics that are most relevant. Hence career choices are a matter of moral judgment. In contrast, from the deontological perspective, this view is considered that you are making yourself an instrument of the betterment of the world and, as it was stated by Kant, no one should be considered as a means to some purpose, but as autonomous ends in themselves.

Effective altruism: A movement aiming to benefit others through the most efficient methods possible.

11. ANTICIPATING RISK IN SCIENCE AND ENGINEERING

Risk in science and engineering: Risk decision in these disciplines is done in multiple ways, such as (1) identifying and assessing risks in their own's work; (2) providing information that is then used to manage risk; and (3) creating new risks.

Qualitative notion of risk: A definition of risk without numerical values or degrees.

Qualitative definitions of risk:

Risk (i): An unwanted event which may occur or not occur.

Risk (ii): The cause of an unwanted event which may or may not occur.

Risk(iii): The spread of possible (good and bad) outcomes that might occur.

Quantitative notion of risk: A definition of risk with numerical values or degree.

Quantitative definitions of risk:

Risk (iv): The probability of an unwanted event which may or may not occur.

Risk (v): The severity of an unwanted event which may or may not occur.

Perception of risks:

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Risk wanted & unwanted: If considered as an outcome it is always unwanted, however if only the cause is considered and its consequential outcomes can be both good and bad it can be a wanted risk. (e.g. smoking per pleasure even though another (negative) outcome can be lung cancer).

Risk – loving: People who put more weight on potential gains than potential losses. People may be risk-loving with respect to some choices and risk-adverse with respect to others.

Risk – adverse: People who put more weight on potential losses than potential gains. People may be risk-loving with respect to some choices and risk-adverse with respect to others.

Leading with risks:

Primary risk prevention: Elimination of a cause of an unwanted event. It links to the definition (i) of risk.

Secondary risk prevention: Finding a means to mitigate the consequences of such a primary risk. It links to the definition (ii) of risk.

Risk assessment: Collecting and assessing the relevant information about risks and characterizing the nature and magnitude of the risk based on this information. It concerns the "value free science".

Risk management: Taking the risk assessment and combining it with economic, technological political and social consideration. It concerns the policy making and "value laden".

> **NOTE:** Eurocodes lacks the ability to separate risk management and risk assessment while the expected value models offer it.

Objective notion of probability: Frequency of which a certain property or event occurs in a population.

Subjective probability: Quantification of risk as the subjective estimate of an individual or maybe a group.

Objective quantification of severity: It refers to the situation in which all the outcomes are in terms of one dimension of value (e.g. all of them in monetary units).

Subjective quantification of severity: It refers to the situation in which the outcomes are in different dimensions (e.g. some in monetary units and others in number of people).

Decision making regarding how much data is available.

Decision making under certainty: Involves cases where an action has only one possible outcome. Therefore risk plays no role.

Decision making under risk: Involves cases where action has more than one possible outcome; results of each outcome is known; probability of each outcome is known.

Decision making under ignorance: Involves cases where action has more than one possible outcome; each outcome is known; probabilities of some outcomes are not known. In these situations both principle of insufficient reason and minimax are the most used procedures.

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Principle of insufficient reason: When one does not know the probabilities of outcomes, each outcome is assigned the same probability.

Minimax (maximin): Choosing the action where the worst possible outcome is the least bad, compared to the other action.

Decision making under deep uncertainty: Action has more than one possible outcome; not all of these outcomes are known, nor are their probabilities. One possibility on how to act in these situations is the use of safety factors. Also precautionary principle applies here.

Safety factors: Characterize the margin of error that is allowed, but how probable the error itself is remains (typically) undetermined. They are rules of thumb, we can incorporate qualitative notion of risk and then make the safety margin larger. The functionality is as follows: (1) We know which is the maximum intended load that a certain object must withstand in service (e.g load), (2) We specify how many times over we want the object to withstand the maximum intended load (e.g twice) and we stablish the safety factor accordingly (e.g 2).

Reasons for using safety factors: (i) Higher loads than those foreseen, (ii) Worse properties of the material than foreseen, (iii) Imperfect theory of the failure mechanism in question, (iv) Possibly unknown failure mechanisms, (v) Human error. Reasons (i), (ii) are quantifiable while (iii), (iv) and (v) are not.

Precautionary principle: If an action or policy has the suspected risk of causing harm, in the absence of scientific consensus whether this action is harmful or not, the burden of proof that it is not harmful in the way that some people claim, lies with those that intend to take action or to propose to perform this particular action or policy. Problems here are: (i) How small suspicion is sufficient?, (ii) What counts as scientific consensus?, (iii) How does one prove that something is not harmful?, (iv) how do we apply it without becoming paralyzed?

Known knowns: Things that we know we know.

Known unknowns: Things that we know that we do not know.

Unknown unknowns: Not even know that we do not know something, therefore wrongly categorize the decision problem.

Expected value of a discrete random variable: Probability weighted average if all its possible values.

Expected value models of risk: Models of risk with a quantifiable probability and a quantifiable severity of each outcome. Whether we want to choose the minimum or maximum value depends on the "currency" we are measuring (e.g. maximize profit vs. minimize fatalities). These models for characterizing and assessing risk, unlike the use of safety factors, allows for optimization. A problem which arises with this model is that there are cases where the expected value for two alternatives might be the same, thus this model is not valid anymore to characterize risk. It is valid as a first step, but then further measures, such as variance, are needed to supplement it.

Utility: The value that something brings to a person.

Expected utility model: Instead of maximizing/minimizing objective values, we maximize/minimize subjective utility.

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Marginal diminishing utility curve: The utility of each additional positive outcome (e.g. dollar) decreases. It refers to the St. Petersburg paradox.

Subjective dimensions of risk: The real-life risk assessment of humans, which may or may not correspond to actual risk. For instance, (i) involuntary risks are generally considered as worse as voluntary ones; (ii) risks where one lacks control as seen as worse; (iii) large risks are sometimes underestimated whereas small risks are sometimes overestimated; (iv) small probability and very bad consequences is seen as worse than high probability and smaller negative consequences.

Arguments for qualitative risk models: The two main reasons are (i) lack of accuracy and (ii) issue of security. If the accuracy of the quantitative model is not sufficiently high it is preferable to develop a structural design that is robust or precautionary or meets the precautionary considerations rather than the one that is economical optimized but potentially has been calculated from inaccurate or unsubstantiated probabilities. Other minor arguments are: (iii) Things need to get informed fast, (iv) computational costs need to be kept low, (v) simplicity.

Utilitarianism: Particular version of consequentialism which says that the moral value of an action is determined exclusively by the moral value of its outcome. Utilitarian's approach to risk are:

Actualist approach: One should assign to an action with uncertain outcomes the utility of the outcome that actually materializes. According to this approach, the actual outcome is what matters, not the possible alternative outcomes that could have happened but did not happen.

The expected utility approach: The morality of uncertain situations can be determined by the expected utility model. This approach faces two main problems: (i) we morally should avoid disproportionally large disasters, which is not that obvious if only expected value is considered and (ii) some people may be exposed unfairly if only expected value if used. In general, they prohibit what might be morally reasonable concerning risk taking and risk distributions.

Deontological approach to risk: It looks into the uncertain possibility that someone's right might be violated (Nozick's crituique). There are three kind of stakeholders here: (i) those who are exposed to the risk, (ii) those who decide how the risk is distributed and (iii) those who benefit from the risk. Thus, whether someone's rights are violated or not lies on some conditions that are related to how many of these stakeholders are the same person. This is a still non-fully answered question and constitutes the main void that this approach has. Possible deontological approaches for dealing with risk management are voluntary approach and intention approach.

Nozick's critique: No matter how small the probability of violating your right is, are your rights violated? Or is there some point where the probability is that small that the rights are not violated anymore?

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